

Data Structures and Algorithms

Lecture 1: Introduction

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ADTs

Data Structures

Algorithms

Time & Space Complexity

Roadmap

1. Data Types

How Python
classifies values

2. ADT

The logical blueprint

3. Algorithms

Step-by-step
problem solving

4. Data Structures

Concrete
implementations

5. Complexity

Time vs. space trade-
offs

Data Types

“An **attribute** of data that tells the interpreter how to classify a variable.”

Numeric

``int`, `float`, `complex``

Sequential

``str`, `list`, `tuple``

Booleans

``True`, `False``

Dictionary

Key-value mapping with ``dict``

ADT vs. Data Structure

Abstract Data Type

- The **what**
- Defines allowed operations
- No concrete implementation
- Works as a logical blueprint

Data Structure

- The **how**
- Stores and organizes data
- Concrete implementation
- Controls performance costs

Example View

Aspect	ADT	Data Structure
Role	Interface	Implementation
Stack	push, pop, peek	Array-based stack
Focus	Rules and operations	Storage and efficiency

Algorithms

“ A set of **step-by-step instructions** used to solve a problem. ”

Recipe Analogy

- Recipe = algorithm
- Steps = instructions
- Meal = output

Omelet Example

1. Crack eggs into a bowl
2. Melt butter over medium-low heat
3. Add eggs and cook gently
4. Add filling when eggs begin to set
5. Fold and serve

Data Structures + Algorithms

Data Structures

Store data in a form that supports access and updates.

Algorithms

Process that data to produce results.

Together

They determine correctness, speed, and memory use.

Practical Goal

Pick the right structure before optimizing the code path.

Time & Space Complexity

Time Complexity

How long an algorithm takes to run.

Priority often matters most in production.

Space Complexity

How much memory an algorithm uses.

Trade-off sometimes we use more space to save time.

Rule of thumb: optimize time first unless the environment is tightly memory-constrained.

Key Takeaways

1. Data Types

Python classifies values into categories.

2. ADT

Defines the blueprint and allowed operations.

3. Data Structure

Implements the ADT in a concrete way.

4. Algorithms

Turn inputs into outputs through steps.

Questions?



Next: Python basics and data type operations

References

- Source: [Lecture 1 Introduction](#)
- Related wiki pages:
 - [\[\[python-for-data-structures|Python for Data Structures\]\]](#)
 - [\[\[linked-list|Linked List\]\]](#)
 - [\[\[stack|Stack\]\]](#)